

1. Questions

Two quadratic equations are given below. Solve both of them and answer the question based on it.

I). $x^2 - 3nx + 15n + 3 = 0$

II). $y^2 - (30-m)y + pm = 1$

The product of the roots of equation I is 108, and the ratio of two roots of equation II is 1:2. If the value of p is (n+4), then find the value of (Pn - m)?

- a. 60
- b. 68
- c. 54
- d. 72
- e. None of these

2. Questions

Two quadratic equations are given below. Solve both of them and answer the question based on it.

I). $x^2 - Ax - 72A = 0$

II). $y^2 + By - 112B = 0$

The value of A is 200% more than that of B, and one of the roots of equation I is 24.

Establish the relationship between the smaller roots of both the equations.

- a. $x > y$
- b. $x < y$
- c. $x \geq y$
- d. $x \leq y$
- e. $x = y$

3. Questions

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Three quadratic equations are given below. Solve both of them and answer the question based on it.

I: $3x^2 - (2A+23)x + 228 = 0$

II: $5y^2 - (3B+6)y + 81 = 0$

III: $2z^2 - (2C - 9)z + 117 = 0$

If one of the roots of equations I, II, and III is 12, 3, and 9, respectively, then establish the correct relationship between A, B, and C.

- a. $A > B > C$
- b. $A < B > C$
- c. $A > B < C$
- d. $A > B = C$
- e. $A = B < C$

4. Questions

Two equations are given below. Solve both of them and answer the questions based on it.

I). $Px^2 - 12x - (6Q+3) = 0$

II). $0.4Py^2 - 9y - 42 = 0$

If one of the roots of equation II is 3.5 and the value of Q is 6.66% more than that of P, then find the sum of the larger root of both equations.

- a. 6.5
- b. 8.5
- c. 7.5
- d. 5.5
- e. None of these

5. Questions

Two equations are given below. Solve both of them and answer the questions based on it.

I). $(x-2)^2 = [y^2 - (x+4)]$

II). $(2y-x)^3 = (y-1)^3$

Find the sum of x and y.

- a. 15
- b. 11
- c. 17
- d. 13
- e. None of these

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6. Questions

Three equations are given in column A, and three conditions are given in column B.

Column A	Column B
I: $x^2 + 3x - 88 = 0$	P: The Difference between the roots of the equation is more than 40.
II: $x^2 - 30x + 216 = 0$	Q: The Difference between roots of the equation is 19.
III: $x^2 - x - 462 = 0$	R: larger root is 50% more than the smaller root.

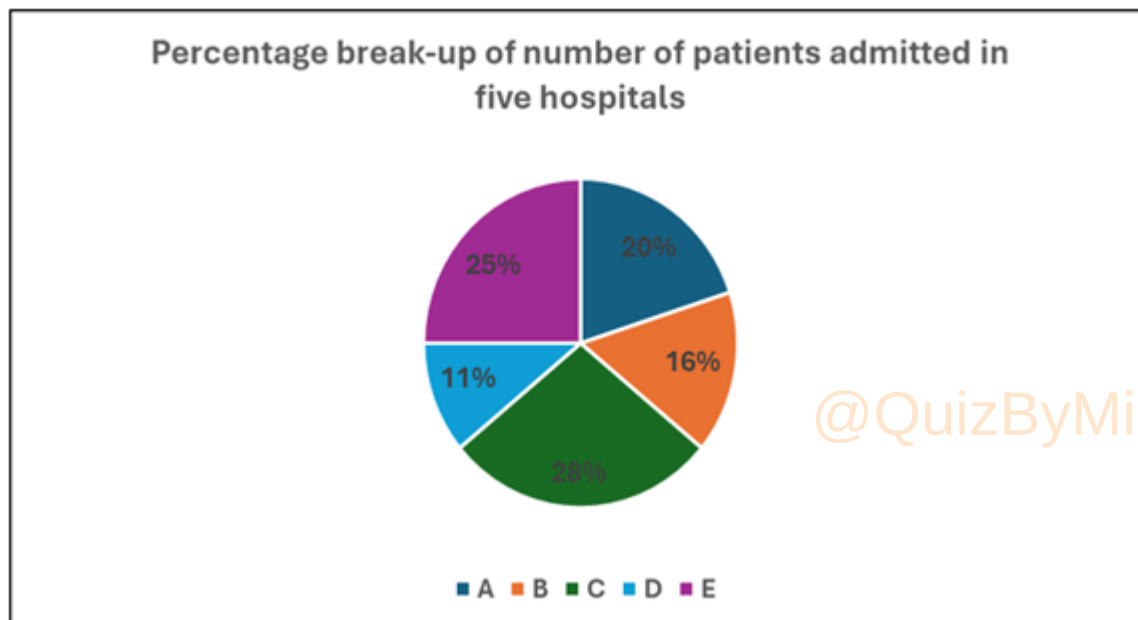
Which among the following options satisfy the given condition?

- I:R, II:Q and III:P
- I:Q, II:R and III: P
- I:Q, II:P and III: R
- I:P, II:R and III:Q
- None of these

7. Questions

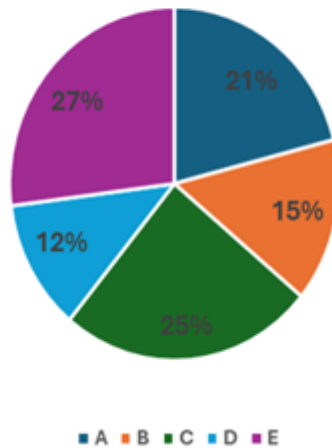
Read the following information carefully and answer the questions based on it-

Two pie charts are given below. The first pie chart represents the percentage breakup of the number of patients admitted in five different hospitals, A, B, C, D, and E, in all hospitals together. The second pie chart represents the percentage breakup of the number of patients recovered in five hospitals together.



QQ

Percentage break-up of number of patients recovered in five hospitals



Notes:

- i). The number of patients admitted in each hospital = the number of patients recovered in each hospital + the number of patients who died in each hospital
- ii). The number of patients who died in hospital D is 25% of the total number of patients admitted in that hospital. The difference between the number of patients admitted in hospitals C and E is 344 less than the number of patients who died in hospital B.

Out of the total number of patients who died in hospital C, 37.5% were above 60 years of age, and (P+Q) patients who died were below 20 years of age. If the number of patients who died between the ages of 20 and 60 years is 27.27% of the total number of patients recovered in hospital C and the value of Q is 50% more than P, then find the value of (2P – Q).

- a. 59
- b. 64
- c. 51
- d. 53
- e. None of these

8. Questions

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The difference between the number of patients recovered in hospitals A and B is “X” and the difference between the number of patients admitted in hospitals B and D is “Y”. Find the difference between the value of Y and X.

- a. 124
- b. 112
- c. 136
- d. 104

e. None of these

9. Questions

In hospital E, one doctor is required for every group of “P” number of patients admitted in the hospital, and there are a total of “O” number of doctors available in the hospital. If the hospital needs 11.11% more doctors and the difference between “P” and the total number of doctors required in the hospital is 14, then find the value of O is approximately what percent of P?

- a. 65%
- b. 72%
- c. 70%
- d. 68%
- e. None of these

10. Questions

Out of the total number of patients admitted to hospital A, 40% were injured in accidents, and the remaining were suffering from diseases. If 25% of patients in hospital A died of any kind of disease, then find the number of people who recovered after accident.

- a. 510
- b. 490
- c. 460
- d. 540
- e. None of these

11. Questions

In another hospital F, the number of patients who died is 45.45% more than that of hospital D, and the average number of patients recovered in hospital D and F is 1288. If the total number of patients admitted in hospital F is $(W^3 - 165)$, then find which among the following statements is/are true regarding “W”.

I). W is one of the factors of the number of patients who died in hospital B

II). W is an even number.

III). $(W+108)$ is a perfect square.

- a. Only I
- b. Only II
- c. Both I and III
- d. Only III

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e. All are incorrect

12. Questions

Out of the total number of patients recovered in hospital B, 45.45% are females. If the number of female patients admitted in hospital B is 192 more than that of males, then find the difference between the number of male and female patients who died in hospital B.

- a. 312
- b. 304
- c. 296
- d. 324
- e. None of these

13. Questions

Read the following data carefully and answer the questions based on it-

The data given below is related to the number of items (TVs, laptops and mobiles) manufactured by five different companies (M, N, O, P and Q) in January. The table given below shows the difference between the number of TVs and laptops manufactured by each company and the average number of laptops and mobiles manufactured by each company. The percentage of items remaining unsold in each company is also given. The number of laptops manufactured by each company is the lowest.

Company	The difference between number of TVs and laptops manufactured	The average number of laptops and mobiles manufactured	% of items remain unsold
M	-	$2z+1$	$(z+5)\%$
N	Y	$3y+7.5$	-
O	$(3z/5) + 3$	-	$(y-5)\%$
P	-	$3.75y$	$1.5x\%$
Q	$(y+2)/2$	$2y+12.5$	-

Notes:

I). Some values are missing in the table.

II). $y^2 - x^2 = 324$ and $x + y = 54$

III). $y + z = 65$

IV). $2z + 2 = 3x$

V). The number of items sold by company N is 225, which is 71.42% of the total number of items manufactured by company N, and the ratio of the total number of items manufactured by companies N

and P is 9:10, respectively.

VI). The Difference between the number of mobiles and laptops manufactured by company Q is 17.

The average number of TVs manufactured by companies P and M is 111.5; then, the total number of items sold by company M is approximately what percent more or less than the number of items sold by company P?

- a. 42%
- b. 32%
- c. 28%
- d. 36%
- e. None of these

14. Questions

Out of the total number of items sold by company N, 44.44% were TVs. Out of the total number of laptops manufactured by company N, 33.33% remain unsold. Find the difference between the number of mobiles sold and unsold by company N.

- a. 35
- b. 25
- c. 30
- d. 20
- e. None of these

15. Questions

The ratio of laptops manufactured by companies O and P is 28:25, and the sum of the number of mobiles manufactured by companies O and P is 238. If the number of TVs manufactured by company O is $(x + 4)$ more than that of company Q, then find the average number of items manufactured by companies O and P together?

- a. 350
- b. 325
- c. 315
- d. 305
- e. None of these

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16. Questions

The average number of TVs and laptops manufactured by company N is what percent of the difference between the total number of items manufactured by companies P and Q?

- a. $3(x+4)\%$
- b. $(y-3)4\%$
- c. $(z+6)\%$
- d. Both A and C
- e. None of these

17. Questions

Out of the total number of items sold by company Q, 10% items were defective, and out of the total number of items remaining unsold by company Q, 14.28% items were defective. If the total number of defective items manufactured by the company is 27, then find the number of non-defective items sold by company Q?

- a. 112
- b. 96
- c. 118
- d. 108
- e. None of these

18. Questions

Read the following information carefully and answer the questions based on it-

Vikas travels some distance by two trains A and B and some distance in car. He travels $(2x+30)$ km distance by train A and $(y-10)$ km distance by train B. The ratio of the lengths of trains A and B is 7:10, respectively. Train A can cross a platform having the length of "1.5x" m in 12 seconds, whereas train B takes 0.6 seconds less than train A to cross the same platform. Trains A and B can cross each other running in opposite directions in $(17D/45)$ seconds, and while running in the same direction, train B crosses train A in 51 seconds. The speed of the car is 33.33% less than the average speed of trains A and B together. He travels $(2x+36)$ km distance with the initial speed of the car and 62.5% more distance with 6km/h more speed. The time taken by Vikas to travel the whole journey is C seconds. The value of D is 10% of the length of train B, and the value of y is $(D^2 + 25)$.

Which among the following values will come in the place of "C"?

- a. 1032
- b. 11210
- c. 55560
- d. 927.5
- e. None of these

19. Questions

Find the time taken by Vikas to travel the whole journey by train A is how much more than the time taken by him to travel the whole journey by train B?

- a. $3(13/40)$ hours
- b. $2(23/40)$ hours
- c. $2(11/20)$ hours
- d. $2(33/40)$ hours
- e. None of these

20. Questions

Find the time taken by train A to completely cross the car running in the same direction.

- a. 25 seconds
- b. 21 seconds
- c. 15 seconds
- d. 23 seconds
- e. None of these

Explanations:

1. Questions

Answer: B

$$I: x^2 - 3nx + 15n + 3 = 0$$

$$\text{Product of roots} = 108$$

$$(15n + 3) = 108$$

$$15n = 105$$

$$n = 7$$

$$x^2 - 21x + 108 = 0$$

$$x^2 - 12x - 9x + 108 = 0$$

$$x(x-12) - 9(x-12) = 0$$

$$x = 12 \text{ or } 9$$

$$p = n + 4 = 7 + 4$$

$$P = 11$$

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$$\text{II: } y^2 - (30 - m)y + pm = 1$$

$$y^2 - (30 - m)y + 11m - 1 = 0$$

Let roots of equation = a and 2a

$$30 - m = 3a$$

$$m = 30 - 3a \text{---(I)}$$

$$11m - 1 = 2a^2$$

$$11(30 - 3a) - 1 = 2a^2$$

$$330 - 33a - 1 = 2a^2$$

$$2a^2 + 33a - 329 = 0$$

$$2a^2 + 47a - 14a - 329 = 0$$

$$2a(a - 7) + 47(a - 7)$$

$$a = -23.5 \text{ or } 7$$

$$a = 7$$

Roots of equation II are 7 and 14

$$m = 30 - 21$$

$$m = 9$$

$$p = 11$$

$$(pn - m) = 11 \times 7 - 9 = 68$$

Hence, the answer is option B.

2. Questions

Answer: B

Let the value of B = m

Value of A = 3m

One root of equation I = 24

$$\text{I: } x^2 - Ax - 72A = 0$$

$$(24)^2 - 72m - 216m = 0$$

$$576 = 288m$$

$$m = 2$$

$$A = 6 \text{ and } B = 2$$

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$$x^2 - 6x - 432 = 0$$

$$x^2 - 24x + 18x - 432 = 0$$

$$x(x-24) + 18(x-24) = 0$$

$$x = +24 \text{ or } -18$$

$$\text{Smaller root} = -18$$

$$\text{II: } y^2 + 2y - 224 = 0$$

$$y^2 + 2y - 224 = 0$$

$$y^2 + 16y - 14y - 224 = 0$$

$$y(y+16) - 14(y+16) = 0$$

$$y = 14 \text{ or } -16$$

$$\text{Smaller root} = -16$$

$$x < y$$

Hence, the answer is option B.

3. Questions

Answer: C

$$\text{I: } 3x^2 - (2A+23)x + 228 = 0$$

$$\text{One root} = 12$$

$$3(12)^2 - (2A+23)12 + 228 = 0$$

$$144 - (2A+23)12 + 228 = 0$$

$$36 - (2A+23) + 19 = 0$$

$$2A+23 = 55$$

$$A = 16$$

$$\text{II: } 5y^2 - (3B+6)y + 81 = 0$$

$$\text{One root} = 3$$

$$5(3)^2 - (3B+6)3 + 81 = 0$$

$$5 - (B+2) + 9 = 0$$

$$B+2 = 14$$

$$B = 12$$

$$\text{III: } 2z^2 - (2C-9)z + 117 = 0$$

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One root = 9

$$2(9)^2 - (2C - 9)9 + 117 = 0$$

$$18 - (2C - 9) + 13 = 0$$

$$2C - 9 = 31$$

$$C = 20$$

$$A > B < C$$

Hence, the answer is option C.

4. Questions

Answer: A

$$\text{II: } 0.4Py^2 - 9y - 42 = 0$$

$$\text{One root} = 3.5$$

$$0.4P(49/4) - 9(7/2) - 42 = 0$$

$$4.9P = 73.5$$

$$P = 15$$

$$Q = 16$$

$$6y^2 - 9y - 42 = 0$$

$$6y^2 - 21y + 12y - 42 = 0$$

$$6y(y + 2) - 21(y + 2) = 0$$

$$y = -2 \text{ or } 3.5$$

$$\text{I: } Px^2 - 12x - (6Q + 3) = 0$$

$$15x^2 - 12x - 99 = 0$$

$$15x^2 - 45x + 33x - 99 = 0$$

$$15x(x - 3) + 33(x - 3) = 0$$

$$x = -2.2 \text{ or } +3$$

$$\text{Sum of larger root of both equations} = 3 + 3.5 = 6.5$$

Hence, the answer is option A.

5. Questions

Answer: D

$$\text{II: } (2y - x)^3 = (y - 1)^3$$

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$$2y - x = y - 1$$

$$x - y = 1$$

$$y = x - 1 \text{ ---(I)}$$

$$\text{I: } (x-2)^2 = [y^2 - (x+4)]$$

$$x^2 - 4x + 4 = y^2 - x - 4$$

$$x^2 - 3x - (x-1)^2 + 8 = 0$$

$$x^2 - 3x - x^2 - 1 + 2x + 8 = 0$$

$$x = 7$$

$$y = 6$$

$$x + y = 13$$

Hence, the answer is option D.

6. Questions

Answer: B

$$\text{I: } x^2 + 3x - 88 = 0$$

$$x^2 + 11x - 8x - 88 = 0$$

$$x(x+11) - 8(x+11) = 0$$

$$x = -11 \text{ or } 8$$

$$\text{II: } x^2 - 30x + 216 = 0$$

$$x^2 - 18x - 12x + 216 = 0$$

$$x(x-18) - 12(x-18) = 0$$

$$x = 12 \text{ or } 18$$

$$\text{III: } x^2 - x - 462 = 0$$

$$x^2 - 22x + 21x - 462 = 0$$

$$x(x-22) + 21(x-22) = 0$$

$$x = -21 \text{ or } 22$$

P: The Difference between the roots of the equation is more than 40.

From III

$$22 - (-21) = 43$$

This is satisfied.

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Q: The Difference between roots of the equation is 19.

From I

$$8 - (11) = 19$$

This is satisfied.

R: larger root is 50% more than the smaller root.

From II

$$[(18 - 12)/12] * 100 = 50\%$$

This is satisfied.

I: Q, II: R and III: P

Hence, the answer is option B.

7. Questions

Let the total number of patients admitted in all hospitals together = $1000x$

Number of patients admitted in hospital D = 11% of $1000x = 110x$

Number of patients who died in hospital D = 25% of $110x = 27.5x$

Number of patients recovered in hospital D = $110x - 27.5x = 82.5x$

Let the number of patients recovered in all hospitals together = $(82.5x/12) * 100 = 687.5x$

Number of patients admitted in hospital B = 16% of $1000x = 160x$

Number of patients recovered in hospital B = 15% of $687.5x = 103.125x$

Number of patients who died in hospital B = $160x - 103.125x = 56.875x$

Number of patients admitted in hospital C = 28% of $1000x = 280x$

Number of patients admitted in hospital E = 25% of $1000x = 250x$

$$280x - 250x = 56.875x - 344$$

$$26.875x = 344$$

$$x = 12.8$$

Total number of patients admitted in all hospitals together = 12800

Total number of patients recovered in all hospitals together = 8800

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Hospital	Number of patients admitted	Number of patients recovered	Number of patients who died
A	2560	1848	712
B	2048	1320	728
C	3584	2200	1384
D	1408	1056	352
E	3200	2376	824

Answer: D

Total number of patients who died in hospital C = 1384

Number of patients who died who were above 60 years of age = $1384 \times \frac{3}{8} = 519$

Number of patients who died between the age of 20 and 60 years = $2200 \times \frac{3}{11} = 600$

Number of patients who died below the age of 20 years = $1384 - (519 + 600) = 265$

$P + Q = 265$

$P = 2a$

$Q = 3a$

$P = (265/5) \times 2 = 106$

$Q = (265/5) \times 3 = 159$

$2P - Q = 106 \times 2 - 159 = 53$

Hence, the answer is option D.

8. Questions

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Answer: B

The difference between the number of patients recovered in hospital A and B = "X"

$$X = 1848 - 1320 = 528$$

The difference between the number of patients admitted in hospital B and D = "Y"

$$Y = 2048 - 1408 = 640$$

$$Y - X = 640 - 528 = 112$$

Hence, the answer is option B.

9. Questions

Let the total number of patients admitted in all hospitals together = 1000x

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Number of patients recovered in hospital D = 110x - 27.5x = 82.5x

Let the number of patients recovered in all hospitals together = $(82.5x/12) * 100 = 687.5x$

Number of patients admitted in hospital B = 16% of 1000x = 160x

Number of patients recovered in hospital B = 15% of 687.5x = 103.125x

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Answer: C

Total number of patients in hospital E = 3200

let the number of doctors in hospital = $O = 9x$

Total number of doctors required = $9x * 10/9 = 10x$

$P = 10x + 14$ or $10x - 14$

If $P = 10x + 14$

$$3200/10x = 10x + 14$$

$$320 = 10x^2 + 14x$$

$$5x^2 + 7x - 160 = 0$$

$$5x^2 - 25x + 32x - 160 = 0$$

$$5x(x-5) + 32(x-5)$$

$$x = -32/5 \text{ or } 5$$

$$x = 5$$

$$P = 64$$

$$O = 45$$

If $P = 10x - 14$

$$320 = 10x^2 - 14x$$

$$5x^2 - 7x - 160 = 0$$

$$5x^2 - 32x + 25x - 160 = 0$$

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$$5x(x+5)-32(x+5)=0$$

$$x = 32/5 \text{ or } -5$$

$$x = 6.4$$

$$P = 50$$

$$O = 57.6 \text{ (not possible)}$$

So the required percentage

$$45/64 * 100$$

$$70.31\% \sim 70\%$$

Hence, the answer is option C.

10. Questions

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Answer: B

Total number of patients admitted in hospital A= 2560

Number of patients injured in accidents = $2560 \times 40\% = 1024$

Number of patients who died due to accident = $(712/4) \times 3 = 534$

Number of patients recovered after accident = $1024 - 534 = 490$

Hence, the answer is option B

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Answer: C

Total number of patients who died in hospital F = $352 \times \frac{16}{11} = 512$

Number of patients recovered in hospital F = $2 \times 1288 - 1056 = 1520$

Total number of patients admitted in hospital F = $1520 + 512 = 2032$

$$W^3 - 165 = 2032$$

$$W^3 = 2197$$

$$W = 13$$

$$I: 728/13 = 56 \text{ (True)}$$

II: 13 is not an even number

$$III: 13 + 108 = 121 \text{ (True)}$$

Hence, the answer is option C.

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Number of patients admitted in hospital B = 16% of $1000x = 160x$

Number of patients recovered in hospital B = 15% of $687.5x = 103.125x$

Number of patients who died in hospital B = $160x - 103.125x = 56.875x$

Number of patients admitted in hospital C = 28% of $1000x = 280x$

Number of patients admitted in hospital E = 25% of $1000x = 250x$

$$280x - 250x = 56.875x - 344$$

$$26.875x = 344$$

$$x = 12.8$$

Total number of patients admitted in all hospitals together = 12800

Total number of patients recovered in all hospitals together = 8800

Hospital	Number of patients admitted	Number of patients recovered	Number of patients who died
A	2560	1848	712
B	2048	1320	728
C	3584	2200	1384
D	1408	1056	352
E	3200	2376	824

Answer: A

Total number of patients admitted to hospital B = 2048

Number of female patients admitted = $(2048 + 192)/2 = 1120$

Number of male patients admitted = $2048 - 1120 = 928$

Number of female patients recovered = $1320 \times 5/11 = 600$

Number of female patients who died = $1120 - 600 = 520$

Number of male patients who died = $728 - 520 = 208$

Required difference = $520 - 208 = 312$

Hence, the answer is option A.

13. Questions

$$y^2 - x^2 = 324$$

$$(y - x)(x + y) = 324$$

$$y - x = 6 \rightarrow (1)$$

WKT

$$x + y = 54 \rightarrow (2)$$

Solve (1) and (2)

$$2y = 60$$

$$y = 30$$

From (1)

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$$30 - x = 6$$

$$x = 24$$

Let

$$y + z = 65$$

$$30 + z = 65$$

$$z = 35$$

Company	The difference between number of TVs and laptops manufactured	The average number of laptops and mobiles manufactured	% of items remain unsold
M	-	71	40%
N	30	97.5	-
O	24	-	25%
P	-	112.5	36%
Q	16	72.5	-

Total number of items sold by company N = 225

Number of items manufactured by company N = $(225/71.42) * 100 = (225/5) * 7 = 315$

Number of items remaining unsold by company N = $315 - 225 = 90$

Number of laptops and mobiles manufactured by company N = $97.5 * 2 = 195$

Number of TVs manufactured by company N = $315 - 195 = 120$

Number of laptops manufactured by company N = $120 - 30 = 90$

Number of mobiles manufactured by company N = $315 - 120 - 90 = 105$

Number of items manufactured by company P = $315/9 * 10 = 350$

Number of items unsold by company P = $36\% \text{ of } 350 = 126$

Number of items sold by company P = $350 - 126 = 224$

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Number of laptops + mobiles manufactured by company P = 225

Number of TVs manufactured by company P = $350 - 225 = 125$

Number of mobiles and laptops manufactured by company Q = $72.5 * 2 = 145$

The Difference between the number of mobiles and laptops manufactured by company Q is 17.

Number of mobiles manufactured by company Q = $(145 + 17)/2 = 81$

Number of laptops manufactured by company Q = $(145 - 17)/2 = 64$

Number of TVs manufactured by company Q = $64 + 16 = 80$

Company	Number of items manufactured	Number of TVs manufactured	Number of laptops manufactured	Number of mobiles manufactured	Number of items sold
M	-	-	-	-	60%
N	315	120	90	105	225
O	-	-	-	-	75%
P	350	125	-	-	224
Q	225	80	64	81	-

Answer: D

According to the question,

Number of TVs manufactured by companies M and P = 223

Number of TVs manufactured by company M = 223 - 125 = 98

Total number of items manufactured by company M = 98 + 71 * 2 = 240

Number of items sold by company M = 240 * 60% = 144

Number of items sold by company P = 224

Thus, percentage = $(224 - 144) / 224 * 100$

= 35.71% ~ 36%

Hence, the answer is option D.

14. Questions

$$y^2 - x^2 = 324$$

$$(y - x)(x + y) = 324$$

$$y - x = 6 \rightarrow (1)$$

WKT

$$x + y = 54 \rightarrow (2)$$

Solve (1) and (2)

$$2y = 60$$

$$y = 30$$

From (1)

$$30 - x = 6$$

$$x = 24$$

Let

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$$y + z = 65$$

$$30 + z = 65$$

$$z = 35$$

Company	The difference between number of TVs and laptops manufactured	The average number of laptops and mobiles manufactured	% of items remain unsold
M	-	71	40%
N	30	97.5	-
O	24	-	25%
P	-	112.5	36%
Q	16	72.5	-

Total number of items sold by company N = 225

Number of items manufactured by company N = $(225/71.42) * 100 = (225/5) * 7 = 315$

Number of items remaining unsold by company N = $315 - 225 = 90$

Number of laptops and mobiles manufactured by company N = $97.5 * 2 = 195$

Number of TVs manufactured by company N = $315 - 195 = 120$

Number of laptops manufactured by company N = $120 - 30 = 90$

Number of mobiles manufactured by company N = $315 - 120 - 90 = 105$

Number of items manufactured by company P = $315/9 * 10 = 350$

Number of items unsold by company P = $36\% \text{ of } 350 = 126$

Number of items sold by company P = $350 - 126 = 224$

Number of laptops + mobiles manufactured by company P = 225

Number of TVs manufactured by company P = $350 - 225 = 125$

Number of mobiles and laptops manufactured by company Q = $72.5 * 2 = 145$

The Difference between the number of mobiles and laptops manufactured by company Q is 17.

Number of mobiles manufactured by company Q = $(145 + 17)/2 = 81$

Number of laptops manufactured by company Q = $(145 - 17)/2 = 64$

Number of TVs manufactured by company Q = $64 + 16 = 80$

Company	Number of items manufactured	Number of TVs manufactured	Number of laptops manufactured	Number of mobiles manufactured	Number of items sold
M	-	-	-	-	60%
N	315	120	90	105	225
O	-	-	-	-	75%
P	350	125	-	-	224
Q	225	80	64	81	-

Answer: B

Number of TVs sold by company N = $225/9 \times 4 = 100$

Number of laptops sold by company N = $90 \times 2/3 = 60$

Number of mobiles sold by company N = $225 - 160 = 65$

Number of mobiles unsold by company N = $105 - 65 = 40$

Required difference = $65 - 40 = 25$

Hence, the answer is option B.

15. Questions

$$y^2 - x^2 = 324$$

$$(y - x)(x + y) = 324$$

$$y - x = 6 \rightarrow (1)$$

WKT

$$x + y = 54 \rightarrow (2)$$

Solve (1) and (2)

$$2y = 60$$

$$y = 30$$

From (1)

$$30 - x = 6$$

$$x = 24$$

Let

$$y + z = 65$$

$$30 + z = 65$$

$$z = 35$$

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Company	The difference between number of TVs and laptops manufactured	The average number of laptops and mobiles manufactured	% of items remain unsold
M	-	71	40%
N	30	97.5	-
O	24	-	25%
P	-	112.5	36%
Q	16	72.5	-

Total number of items sold by company N = 225

Number of items manufactured by company N = $(225/71.42) * 100 = (225/5) * 7 = 315$

Number of items remaining unsold by company N = $315 - 225 = 90$

Number of laptops and mobiles manufactured by company N = $97.5 * 2 = 195$

Number of TVs manufactured by company N = $315 - 195 = 120$

Number of laptops manufactured by company N = $120 - 30 = 90$

Number of mobiles manufactured by company N = $315 - 120 - 90 = 105$

Number of items manufactured by company P = $315/9*10 = 350$

Number of items unsold by company P = $36\% \text{ of } 350 = 126$

Number of items sold by company P = $350 - 126 = 224$

Number of laptops + mobiles manufactured by company P = 225

Number of TVs manufactured by company P = $350 - 225 = 125$

Number of mobiles and laptops manufactured by company Q = $72.5 * 2 = 145$

The Difference between the number of mobiles and laptops manufactured by company Q is 17.

Number of mobiles manufactured by company Q = $(145 + 17)/2 = 81$

Number of laptops manufactured by company Q = $(145 - 17)/2 = 64$

Number of TVs manufactured by company Q = $64 + 16 = 80$

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Company	Number of items manufactured	Number of TVs manufactured	Number of laptops manufactured	Number of mobiles manufactured	Number of items sold
M	-	-	-	-	60%
N	315	120	90	105	225
O	-	-	-	-	75%
P	350	125	-	-	224
Q	225	80	64	81	-

Answer: C

Let the number of laptops manufactured by company O = $28x$

Number of laptops manufactured by company P = $25x$

The number of TVs manufactured by company O = $80 + 28 = 108$

Number of laptops manufactured by company O = $108 - 24 = 84$

Number of laptops manufactured by company P = $84 * (25/28) = 75$

Number of mobiles manufactured by company P = $225 - 75 = 150$

Number of mobiles manufactured by company O = $238 - 150 = 88$

Number of items manufactured by company P and O together = $350 + 84 + 108 + 88 = 630$

Average number of items manufactured by company P and O together = $630/2 = 315$

Hence, the answer is option C.

16. Questions

$$y^2 - x^2 = 324$$

$$(y - x)(x + y) = 324$$

$$y - x = 6 \rightarrow (1)$$

WKT

$$x + y = 54 \rightarrow (2)$$

Solve (1) and (2)

$$2y = 60$$

$$y = 30$$

From (1)

$$30 - x = 6$$

$$x = 24$$

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Let

$$y + z = 65$$

$$30 + z = 65$$

$$z = 35$$

Company	The difference between number of TVs and laptops manufactured	The average number of laptops and mobiles manufactured	% of items remain unsold
M	-	71	40%
N	30	97.5	-
O	24	-	25%
P	-	112.5	36%
Q	16	72.5	-

Total number of items sold by company N = 225

$$\text{Number of items manufactured by company N} = (225/71.42) * 100 = (225/5) * 7 = 315$$

$$\text{Number of items remaining unsold by company N} = 315 - 225 = 90$$

$$\text{Number of laptops and mobiles manufactured by company N} = 97.5 * 2 = 195$$

$$\text{Number of TVs manufactured by company N} = 315 - 195 = 120$$

$$\text{Number of laptops manufactured by company N} = 120 - 30 = 90$$

$$\text{Number of mobiles manufactured by company N} = 315 - 120 - 90 = 105$$

$$\text{Number of items manufactured by company P} = 315/9 * 10 = 350$$

$$\text{Number of items unsold by company P} = 36\% \text{ of } 350 = 126$$

$$\text{Number of items sold by company P} = 350 - 126 = 224$$

$$\text{Number of laptops + mobiles manufactured by company P} = 225$$

$$\text{Number of TVs manufactured by company P} = 350 - 225 = 125$$

$$\text{Number of mobiles and laptops manufactured by company Q} = 72.5 * 2 = 145$$

The Difference between the number of mobiles and laptops manufactured by company Q is 17.

$$\text{Number of mobiles manufactured by company Q} = (145 + 17)/2 = 81$$

$$\text{Number of laptops manufactured by company Q} = (145 - 17)/2 = 64$$

$$\text{Number of TVs manufactured by company Q} = 64 + 16 = 80$$

Company	Number of items manufactured	Number of TVs manufactured	Number of laptops manufactured	Number of mobiles manufactured	Number of items sold
M	-	-	-	-	60%
N	315	120	90	105	225
O	-	-	-	-	75%
P	350	125	-	-	224
Q	225	80	64	81	-

Answer: A

Average number of TVs and laptops manufactured by company N $= (120 + 90)/2 = 105$

Difference between total number of items manufactured by company P and Q $= 350 - 225 = 125$

Thus, percentage $= (105/125) * 100$

$$84\% = 3(x+4)\%$$

Hence, the answer the option A.

17. Questions

$$y^2 - x^2 = 324$$

$$(y - x)(x+y) = 324$$

$$y - x = 6 \rightarrow (1)$$

WKT

$$x + y = 54 \rightarrow (2)$$

Solve (1) and (2)

$$2y = 60$$

$$y = 30$$

From (1)

$$30 - x = 6$$

$$x = 24$$

Let

$$y + z = 65$$

$$30 + z = 65$$

$$z = 35$$

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Company	The difference between number of TVs and laptops manufactured	The average number of laptops and mobiles manufactured	% of items remain unsold
M	-	71	40%
N	30	97.5	-
O	24	-	25%
P	-	112.5	36%
Q	16	72.5	-

Total number of items sold by company N = 225

Number of items manufactured by company N = $(225/71.42) * 100 = (225/5) * 7 = 315$

Number of items remaining unsold by company N = $315 - 225 = 90$

Number of laptops and mobiles manufactured by company N = $97.5 * 2 = 195$

Number of TVs manufactured by company N = $315 - 195 = 120$

Number of laptops manufactured by company N = $120 - 30 = 90$

Number of mobiles manufactured by company N = $315 - 120 - 90 = 105$

Number of items manufactured by company P = $315/9 * 10 = 350$

Number of items unsold by company P = $36\% \text{ of } 350 = 126$

Number of items sold by company P = $350 - 126 = 224$

Number of laptops + mobiles manufactured by company P = 225

Number of TVs manufactured by company P = $350 - 225 = 125$

Number of mobiles and laptops manufactured by company Q = $72.5 * 2 = 145$

The Difference between the number of mobiles and laptops manufactured by company Q is 17.

Number of mobiles manufactured by company Q = $(145 + 17)/2 = 81$

Number of laptops manufactured by company Q = $(145 - 17)/2 = 64$

Number of TVs manufactured by company Q = $64 + 16 = 80$

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Company	Number of items manufactured	Number of TVs manufactured	Number of laptops manufactured	Number of mobiles manufactured	Number of items sold
M	-	-	-	-	60%
N	315	120	90	105	225
O	-	-	-	-	75%
P	350	125	-	-	224
Q	225	80	64	81	-

Answer: D

Total number of items manufactured by company Q = 225

let number of items sold = a

Number of items unsold = 225 – a

$$a \cdot 10\% + (225 - a) / 7 = 27$$

$$a / 10 + (225 - a) / 7 = 27$$

$$7a + 2250 - 10a = 1890$$

$$3a = 360$$

$$a = 120$$

Number of items sold by company Q = 120

Number of non-defective items sold by company Q = $120 \cdot 90\% = 108$

Hence, the answer is option D.

18. Questions

Let the length of train A = 7a km

The length of train B = 10a km

Length of platform = 1.5x m

The value of D = 10% of 10a = a

Speed of train A = $(7a + 1.5x) / 12$ mps

Speed of train B = $(10a + 1.5x) / 11.4 = (100a + 15x) / 114$ mps

$$[(7a + 1.5x) / 12 + (100a + 15x) / 114] \cdot 17D / 45 = 17a$$

$$[(7a + 1.5x) / 12 + (100a + 15x) / 114] = 45$$

$$(133a + 28.5x + 200a + 30x) / 228 = 45$$

$$333a + 58.5x = 45 \cdot 228$$

$$37a + 6.5x = 1140 \text{ ---(I)}$$

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$$[(100a+15x)/114 - (7a + 1.5x)/12]*51 = 17a$$

$$[(100a + 15x)/38 - (7a + 1.5x)/4] = a$$

$$[200a + 30x - (133a + 28.5x)] = 76a$$

$$67a + 1.5x = 76a$$

$$1.5x = 9a$$

$$x:a = 6m:1m$$

$$37*m + 6.5*6m = 1140$$

$$37m + 39m = 1140$$

$$m = 15$$

$$a = 15$$

$$x = 90$$

$$\text{Length of train A} = 7a = 7 * 15 = 105 \text{ m}$$

$$\text{Length of train B} = 10a = 10 * 15 = 150 \text{ m}$$

$$\text{Length of platform} = 1.5x = 1.5 * 90 = 135 \text{ m}$$

$$\text{Speed of train A} = (105 + 135)/12 = 20 \text{ m/sec} = 20 * 18/5 = 72 \text{ kmph}$$

$$\text{Speed of train B} = (150 + 135)/11.4 = 25 \text{ m/sec} = 25 * 18/5 = 90 \text{ kmph}$$

$$D = 15$$

$$y = 225 + 25 = 250$$

Answer: C

$$\text{Time taken by Vikas to cover } (2x+30) \text{ km journey by train A} = 210/72 = 35/12 \text{ hours.}$$

$$\text{Time taken by Vikas to cover } (y - 10) \text{ km journey by train B} = 240/90 = 8/3 \text{ hours}$$

$$\text{Speed of car} = (72+90)/2 * 2/3 = 54 \text{ kmph}$$

$$\text{Time taken to cover } (2x + 36) \text{ km distance} = 216/54 = 4 \text{ h}$$

$$\text{Time taken to cover } 216 * 13/8 \text{ km distance} = 351/60 = 5.85 \text{ h}$$

$$\text{Time taken by him to travel the whole journey} = C = (35/12) + (8/3) + 4 + 5.85 = (35 + 32 + 48 + 70.2)/12 \\ = (185.2)/12 \text{ hours} = 55650 \text{ seconds}$$

$$C = 55560$$

Hence, the answer is option C.

19. Questions

Let the length of train A = 7a km

The length of train B = 10a km

Length of platform = $1.5x$ m

The value of $D = 10\%$ of $10a = a$

Speed of train A = $(7a+1.5x)/12$ mps

Speed of train B = $(10a + 1.5x)/11.4 = (100a + 15x)/114$ mps

$$[(7a + 1.5x)/12 + (100a+15x)/114]*17D/45 = 17a$$

$$[(7a + 1.5x)/12 + (100a+15x)/114] = 45$$

$$(133a+28.5x+200a + 30x)/228 = 45$$

$$333a + 58.5x = 45*228$$

$$37a + 6.5x = 1140 \text{ ---(I)}$$

$$[(100a+15x)/114 - (7a + 1.5x)/12]*51 = 17a$$

$$[(100a + 15x)/38 - (7a + 1.5x)/4] = a$$

$$[200a + 30x - (133a + 28.5x)] = 76a$$

$$67a + 1.5x = 76a$$

$$1.5x = 9a$$

$$x:a = 6m:1m$$

$$37*m + 6.5*6m = 1140$$

$$37m + 39m = 1140$$

$$m = 15$$

$$a = 15$$

$$x = 90$$

Length of train A = $7a = 7 * 15 = 105$ m

Length of train B = $10a = 10 * 15 = 150$ m

Length of platform = $1.5x = 1.5 * 90 = 135$ m

Speed of train A = $(105 + 135)/12 = 20$ m/sec = $20 * 18/5 = 72$ kmph

Speed of train B = $(150 + 135)/11.4 = 25$ m/sec = $25 * 18/5 = 90$ kmph

$$D = 15$$

$$y = 225 + 25 = 250$$

Time taken by Vikas to cover $(2x+30)$ km journey by train A = $210/72 = 35/12$ hours.

Time taken by Vikas to cover $(y - 10)$ km journey by train B = $240/90 = 8/3$ hours

Speed of car = $(72+90)/2*2/3 = 54$ kmph

Time taken to cover $(2x + 36)$ km distance = $216/54 = 4$ h

Time taken to cover $216 \times \frac{13}{8}$ km distance = $351/60 = 5.85$ h

Answer: D

Total distance covered by Vikas = $240 + 210 + 216 + 351 = 1017$ km

Time taken by train A = $1017/72 = 113/8$ hours

Time taken by train B = $1017/90 = 113/10$ hours

$(113/8 - 113/10) = (565 - 452)/40 = 113/40 = 2(33/40)$ hours

Hence, the answer is option D.

20. Questions

Let the length of train A = $7a$ km

The length of train B = $10a$ km

Length of platform = $1.5x$ m

The value of D = 10% of $10a = a$

Speed of train A = $(7a + 1.5x)/12$ mps

Speed of train B = $(10a + 1.5x)/11.4 = (100a + 15x)/114$ mps

$[(7a + 1.5x)/12 + (100a + 15x)/114] \times 17D/45 = 17a$

$[(7a + 1.5x)/12 + (100a + 15x)/114] = 45$

$(133a + 28.5x + 200a + 30x)/228 = 45$

$333a + 58.5x = 45 \times 228$

$37a + 6.5x = 1140$ —(I)

$[(100a + 15x)/114 - (7a + 1.5x)/12] \times 51 = 17a$

$[(100a + 15x)/38 - (7a + 1.5x)/4] = a$

$[200a + 30x - (133a + 28.5x)] = 76a$

$67a + 1.5x = 76a$

$1.5x = 9a$

$x:a = 6m:1m$

$37 \times m + 6.5 \times 6m = 1140$

$37m + 39m = 1140$

$m = 15$

$a = 15$

$x = 90$

Length of train A = $7a = 7 \times 15 = 105$ m

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Length of train B = $10a = 10 * 15 = 150$ m

Length of platform = $1.5x = 1.5 * 90 = 135$ m

Speed of train A = $(105 + 135)/12 = 20$ m/sec = $20 * 18/5 = 72$ kmph

Speed of train B = $(150 + 135)/11.4 = 25$ m/sec = $25 * 18/5 = 90$ kmph

D = 15

y = $225 + 25 = 250$

Time taken by Vikas to cover $(2x+30)$ km journey by train A = $210/72 = 35/12$ hours.

Time taken by Vikas to cover $(y - 10)$ km journey by train B = $240/90 = 8/3$ hours

Speed of car = $(72+90)/2 * 2/3 = 54$ kmph

Time taken to cover $(2x + 36)$ km distance = $216/54 = 4$ h

Time taken to cover $216 * 13/8$ km distance = $351/60 = 5.85$ h

Answer: B

Length of train A = 105 m

Speed of train A = $72 * 5/18 = 20$ mps

Speed of car = $54 * 5/18 = 15$ mps

The time taken = $105/(20-15) = 21$ seconds.

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